

Computer Vision and Deep Learning for Automated Weed Detection and Species Classification in Crop Fields

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BACHELOR OF TECHNOLOGY
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ABSTRACT

Weed infestation significantly impacts crop yield and agricultural productivity. Manual weed detection and traditional control methods are inefficient and environmentally unsustainable. This academic project investigates the use of computer vision and deep learning techniques for automated weed detection and species classification in crop fields. Field images annotated with weed locations and species labels are used to train deep learning models. Object detection techniques identify weed regions within crop images, while classification models distinguish between different weed species. Transfer learning is applied to enhance model performance and reduce training requirements by utilizing pre-trained convolutional neural networks. Model evaluation is conducted using metrics such as accuracy, precision, recall, and F1-score. Although no real-world deployment is implemented, the study demonstrates the potential of intelligent vision-based systems for future precision agriculture applications.

Internal Guide

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